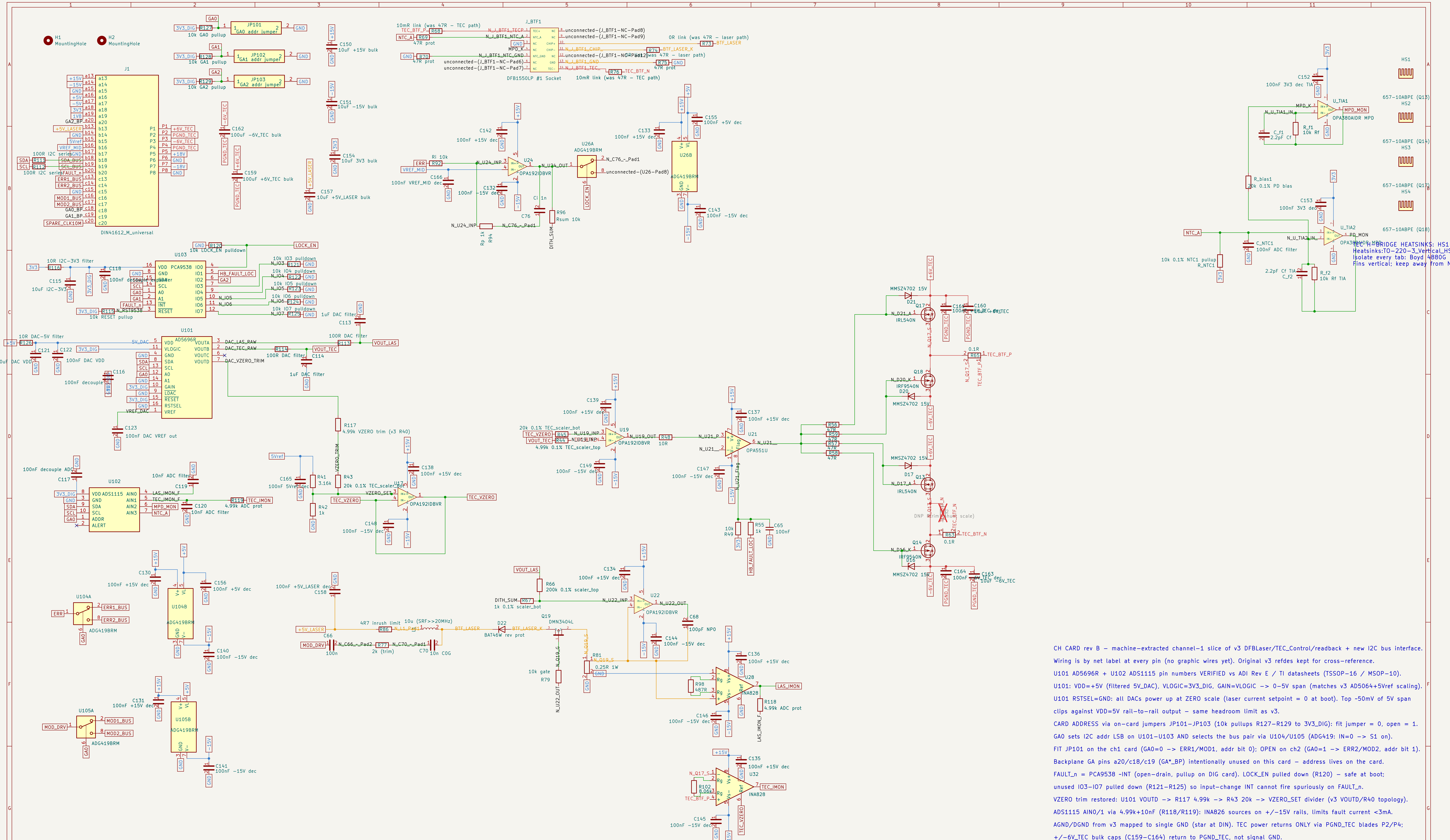

CH CARD rev B — machine-extracted channel-1 slice of v3 DFBLaser/TEC_Control/readback + new I2C bus interface.

Wiring is by net label at every pin (no graphic wires yet). Original v3 refdes kept for cross-reference.

U101 AD5696R + U102 ADS1115 pin numbers VERIFIED vs ADI Rev E / TI datasheets (TSSOP-16 / MSOP-10).

U101: VDD=+5V (filtered 5V_DAC), VLOGIC=3V3_DIG, GAIN=VLOGIC -> 0-5V span (matches v3 AD5064+5Vref scaling).

U101 RSTSEL=GND: all DACs power up at ZERO scale (laser current setpoint = 0 at boot). Top ~50mV of 5V span

clips against VDD=5V rail-to-rail output — same headroom limit as v3.

CARD ADDRESS via on-card jumpers JP101-JP103 (10k pullups R127-R129 to 3V3_DIG): fit jumper = 0, open = 1.

GA0 sets I2C addr LSB on U101-U103 AND selects the bus pair via U104/U105 (ADG419: IN=0 -> S1 on).

FIT JP101 on the ch1 card (GA0=0 -> ERR1/MOD1, addr bit 0); OPEN on ch2 (GA0=1 -> ERR2/MOD2, addr bit 1).

Backplane GA pins a20/c18/c19 (GA*_BP) intentionally unused on this card — address lives on the card.

FAULT_n = PCA9538 ~INT (open-drain, pullup on DIG card). LOCK_EN pulled down (R120) — safe at boot;

unused IO3-IO7 pulled down (R121-R125) so input-change INT cannot fire spuriously on FAULT_n.

VZERO trim restored: U101 VOUTD -> R117 4.99k -> R43 20k -> VZERO_SET divider (v3 VOUTD/R40 topology).

ADS1115 AIN0/1 via 4.99k+10nF (R118/R119): INA826 sources on +/-15V rails, limits fault current <3mA.

AGND/DGND from v3 mapped to single GND (star at DIN). TEC power returns ONLY via PGND_TEC blades P2/P4;

+/-6V_TEC bulk caps (C159-C164) return to PGND_TEC, not signal GND.

Decoupling bank C130-C158: one 100nF per +/-15V IC supply pin + bulk per rail — place at ICs during layout.

OFF-CARD REQUIREMENTS: I2C SDA/SCL + FAULT_n pullups on DIG card.

J_BTF1 butterfly socket = front panel. TEC_VZERO regenerated locally by U17 buffer.

v3 ch1 slice + universal Type M bus interface; review fixes applied (pinouts, 5V DAC, decoupling)

Sheet: /
File: CH_Card.kicad_sch

Title: CH card — laser channel eurocard (rev B, auto-extracted)

Size: A2	Date: 2026-07-16	**Rev:**
KiCad E.D.A. 10.0.3		Id: 1/1